

Mustafa Abdulrahman

mus2003.abdul@gmail.com | linkedin.com/in/mus123/ | github.com/Mustaballer

EDUCATION

University of Toronto

Bachelor of Applied Science in Computer Engineering

Toronto, ON

May 2026

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, C++, SQL, HTML/CSS

Frameworks: React, TanStack Query, Redux, Electron, Playwright, Spring Boot, Django, Flask

AI/Agents: Amazon Bedrock, Strands Agents, LangChain, MCP, Bedrock Guardrails

Cloud/Tools: AWS (S3, IAM, Lambda, EC2, CDK, CloudWatch), Docker, Git, CI/CD, Jenkins, PostgreSQL, MySQL

EXPERIENCE

Software Engineer

May 2026 – Present

Amazon

Toronto, ON

- Owned **full CD** for an **Electron** AI desktop assistant (**Strands**, **Bedrock**) rolled out to **3,000+** finance analysts, building **Playwright** Windows E2E gates and daily prod canaries on **EC2** (CDK) with zero manual approval steps.
- Drove **AppSec** certification of a high-risk **GenAI** agent across **28+** security CRs, fixing **4** threat-model findings and **8** production risks via **Bedrock Guardrails**, per-user **S3/IAM** session isolation, a read-only **SQL** allowlist, and a sandboxed **Python** runtime.
- Resolved **Sev-2** incidents as on-call (silent **Kinesis** record loss) and root-caused a startup deadlock causing **0/18** successful launches by moving serial agent rehydration off a **90s** readiness path into a worker pool.
- Migrated Promise-to-Pay and Dispute workflows on a collections platform used by **~1,000** analysts from **Redux-Saga** to **TanStack Query** (**React**, **TypeScript**), and built group-level tasks for resellers with **100s** of linked accounts.
- Ran **user research** with collections analysts to shape a consolidated multi-account inbox, and partnered with finance teams to build **agentic AI skills** automating receipt matching and cross-country deductions.

Software Engineering Intern

May 2025 – August 2025

Amazon

Toronto, ON

- Designed an end-to-end **GenAI** system (**Bedrock**, **LangChain**, **MCP**) that auto-converts approval policy PDFs into **Java** rules with test cases, reducing turnaround from 2 weeks → 2 days.
- Implemented **agentic workflows** with **TDD** and **LLMOps guardrails** on **Lambda**, **API Gateway**, and **CDK**, eliminating **65%** of potential production rollbacks and saving **~12** engineering hours per policy.

Software Engineering Intern

May 2024 – April 2025

AMD

Markham, ON

- Engineered a system using the **GitHub GraphQL API** to extract and store **2M+** entries in **PostgreSQL** with daily updates, visualized in **Power BI** reports shared weekly with upper management.
- Optimized GPU kernel code in **HIP**, implementing efficient matrix addition and moving average filters, enhancing data processing speed by **35%** and reducing compute time by **20%**.

Machine Learning Engineer Intern

May 2023 – August 2023

MLDSAI Inc.

Toronto, ON

- Developed the **OpenAdapt** Python library for **AI-first process automation**, using **transformer** completions for synthetic input generation to cut manual task input by **50%**.
- Scripted automated recording of user inputs and screenshots, generating a **10,000+** sample dataset and saving **~40%** time over manual collection.
- Accelerated data processing by implementing **tokenization**, reducing processing time by **30%**.

PROJECTS

Hardware Sign-out Site | *React, Redux, Django, PostgreSQL, Docker*

- Directed a **15+ person team** to build an **open-source** inventory system for **5,000+** components, reducing wait times from **4 hours** to **10 minutes** at Canada's largest Makeathon.
- Built an application review portal with **React**, **Django**, and **PostgreSQL**, cutting screening time from **20 hours** to **2 hours** for **1,500+** applicants.